CLASS XI CHEMISTRY — CHAPTER 1

Some Basic Concepts of Chemistry

NEW ADDITION BOOK STYLE • JEE + SCHOOL • ALL MAJOR TOPICS

1. CHEMISTRY & MATTER

Chemistry deals with composition, structure, properties and transformations of matter.

Classification of matter: Pure substances → elements and compounds. Mixtures → homogeneous (uniform) and heterogeneous (non-uniform).

Physical properties: observed without changing composition (melting point, density). **Chemical properties:** describe chemical change (reactivity, combustion).

2. LAWS OF CHEMICAL COMBINATION ■

Law	Key idea
Conservation of mass	Mass is neither created nor destroyed in a chemical reaction.
Definite proportions	A pure compound contains the same elements in a fixed ratio by mass.
Multiple proportions	When two elements form more than one compound, masses of one element that combine with a fixed mass of the
Gay-Lussac's law	Volumes of reacting gases, at same T and P, are in simple whole-number ratios.
Avogadro's law	Equal volumes of gases at same T and P contain equal numbers of molecules.

3. DALTON'S ATOMIC THEORY

Atoms are the basic units of elements; atoms of an element have the same atomic number; compounds form by simple whole-number combinations; chemical reactions involve rearrangement of atoms. Modern view: atoms of an element can have different isotopes, and atoms can be divided into subatomic particles.

4. ATOMIC & MOLECULAR MASSES

Atomic mass unit: 1 u = 1/12 of the mass of one carbon-12 atom. **Relative atomic mass:** weighted average of isotopic masses according to natural abundance.

Average atomic mass = Σ(isotopic mass × fractional abundance)

Molecular mass: sum of atomic masses of all atoms in a molecule. **Formula mass:** used especially for ionic compounds.

5. MOLE CONCEPT ■■■

1 mole contains Avogadro number of entities: **$N_A = 6.022 \times 10^{23}\ mol^{-1}$**.

Moles from number of particles: **$n = N/N_A$**

Moles from mass: **$n = m/M$**, where m = given mass and M = molar mass.

For gases at STP, molar volume is approximately 22.4 L mol^{-1} under the traditional NCERT/STP convention. Always follow the condition stated in the question.

JEE shortcut: Convert every quantity to moles first; use the balanced equation for mole ratios; convert the final answer to the requested unit.

6. MOLAR MASS & PERCENT COMPOSITION

Molar mass = numerical value of molecular/formula mass expressed in g mol^{-1}.

% by mass of an element = (mass of element in 1 mol compound / molar mass of compound) × 100

7. EMPIRICAL & MOLECULAR FORMULA ■■■

Empirical formula: simplest whole-number ratio of atoms. **Molecular formula:** actual number of atoms.

Steps from percentage composition: 1) Assume 100 g. 2) Convert % to grams. 3) Divide each by atomic mass. 4) Divide by the smallest value. 5) Multiply all ratios by 2, 3, 4… if needed to obtain integers.

n = Molecular molar mass / Empirical formula mass; Molecular formula = (Empirical formula) × n

8. STOICHIOMETRY ■■■

A balanced chemical equation gives mole ratios. Example: $2H_2 + O_2 \rightarrow 2H_2O$ means 2 mol H_2 : 1 mol O_2 : 2 mol H_2O.

General workflow: Balance → convert given data to moles → apply coefficient ratio → convert to requested quantity.

Limiting reagent: reactant consumed first. It determines the maximum product. Compare available moles/coefficient for each reactant; the smallest value identifies the limiting reagent.

Excess reagent: remains after the limiting reagent is completely consumed.

% yield = (actual yield / theoretical yield) × 100

9. CONCENTRATION TERMS ■■■

Term	Formula / meaning
Mass %	(mass of solute / mass of solution) × 100
Volume %	(volume of solute / volume of solution) × 100
Mass by volume %	(mass of solute in g / volume of solution in mL) × 100
Molarity (M)	moles of solute / volume of solution in L
Molality (m)	moles of solute / mass of solvent in kg
Mole fraction (χ)	moles of component / total moles of all components
ppm	parts of solute per 10■ parts of solution (often used for very dilute solutions)

Dilution: when only solvent is added, moles of solute remain constant, so $\mathbf{M_1V_1 = M_2V_2}$.

10. SIGNIFICANT FIGURES ■■

All non-zero digits are significant. Zeros between non-zero digits are significant. Leading zeros are not significant. Trailing zeros are significant when a decimal point is explicitly shown.

Multiplication/division: answer has the same number of significant figures as the least precise value. **Addition/subtraction:** answer has the same number of decimal places as the least precise value.

11. SCIENTIFIC NOTATION & DIMENSIONAL ANALYSIS

Write numbers as $a \times 10^n$, with $1 \le a < 10$. In unit conversions, multiply by conversion factors equal to 1 so unwanted units cancel.

12. IMPORTANT JEE FORMULAS / FACTS

- $n = m/M$
- $N = nN_A$
- % composition = (mass of element / molar mass of compound) × 100
- M = moles of solute / volume of solution (L)
- m = moles of solute / mass of solvent (kg)
- $\chi_i = n_i / \Sigma n$
- $M_1V_1 = M_2V_2$
- % yield = actual/theoretical × 100
- At the same T and P, gas volume ∝ number of moles.

13. EXAM CHECKLIST — JEE + SCHOOL

✓ Learn all five laws of chemical combination and their statements.

✓ Be fast with mole ↔ mass ↔ particles conversions.

✓ Practise empirical/molecular formula problems.

✓ Balance equations before doing stoichiometric calculations.

✓ Master limiting reagent and percentage yield.

✓ Know M, m, mole fraction, mass %, volume % and ppm.

✓ Use units carefully and apply significant-figure rules.

✓ For school: write definitions and laws in precise textbook language.

✓ For JEE: practise mixed numerical questions and multi-step conversions.

CONCEPT CLEAR → PRACTICE MORE → SCORE HIGH